

OGUNDEKO AZEEZ ADEBOLA

Prospective Graduate Student | Computer Science & Artificial Intelligence

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RESEARCH INTERESTS

Speech and language technology for low-resource African languages (ASR/TTS, code-switching, Nigerian Pidgin); **trustworthy and verifiable LLM systems** (grounded extraction, citation-backed outputs, separating model proposals from deterministic decisions); **human-in-the-loop autonomous agents** (graduated autonomy, escalation and oversight); and **real-time multimodal AI systems** for voice and vision.

EDUCATION

Lagos State University (LASU), Lagos, Nigeria

Jan 2020 – Nov 2024

B.Sc. Electronics and Computer Engineering — **First Class Honours, CGPA: 4.59 / 5.0**

- National Youth Service Corps (NYSC): completed

RESEARCH & INDEPENDENT PROJECTS

Verifiable LLM Extraction for Financial Compliance (Toyyib) | toyyib.app

2026

- Question:** can LLMs extract facts from long, unstructured financial filings reliably enough for high-stakes decisions without being trusted to make those decisions?
- Designed a pipeline in which the LLM only **proposes typed, citation-backed accounting facts**, while a deterministic rules engine validates them and performs AAOIFI Shariah screening; missing or conflicting evidence marks only dependent screens as insufficient
- Built **full reproducibility**: every result binds filing hashes, accepted observations, prompt/schema hashes and model identity, so any verdict can be re-derived and audited; analyst corrections become scoped, reusable mapping rules
- Evaluated on a real 72-page NGX filing (Docling parsing, OpenAI structured outputs), with per-call and per-run cost limits; work guided by a Shariah advisory board

Speech AI for Nigerian Languages (Vonative) | vonative.com

2025 – Present

- Question:** how do we build low-latency spoken-dialogue agents for low-resource languages, Nigerian Pidgin and code-switched speech?
- Built a real-time voice-agent system (LiveKit, custom dialogue state machine, node-based workflow engine) combining African-language STT/TTS (Spitch, NaijaLingo, ElevenLabs) with LLMs (Groq, Claude) over Nigerian telephony
- Investigated failure modes in deployed agents: **ASR hallucination** on local-language audio, and LLMs **verbalising tool calls** instead of executing them, which I traced to prompt phrasing and weakly typed tool-output schemas
- Designed a multilingual IVR banking dialogue flow covering **five Nigerian languages**

Human-in-the-Loop Autonomous Agents (Sarah AI) | trysarah.ai

2026 – Present

- Designing **graduated autonomy** (Review → Supervised → Autonomous) for AI collections agents: routine actions run autonomously within approved limits, while disputes, vulnerability and complaints escalate to humans, with autonomy increasing per tenant as confidence builds

Real-Time Computer Vision Platform (InvigilX and follow-up design)

2025 – 2026

- Built InvigilX, integrating **computer-vision models** with a live alert pipeline for real-time anomaly detection
- Designed a follow-up platform with a hook-chain execution model (frame → detect → track → condition → alert), per-camera real-time streaming and bring-your-own-compute local inference

Semantic Matching for Scholarship Discovery (ScholarAI)

2025

- Built **embedding-based semantic matching** (pgvector) between parsed student CVs and scholarship/professor data collected by web scraping, with an asynchronous processing pipeline; requirements shaped by research into graduate-student behaviour

Consistent Real-Time State for Multiplayer Systems (Mazamaza) | mazamaza.xyz

2026 – Present

- Designed a **Redis-backed distributed state engine** with atomic operations and fault recovery, plus Cloudflare Durable Objects for per-room state, keeping hundreds of concurrent WebSocket sessions consistent
- Built a synchronised tournament engine (stage ticker, per-room advancement) and six game modes; launched to **100+ users**

PROFESSIONAL EXPERIENCE

Software Engineer – Sarah AI

Sep 2026 – Present

Remote Contract (United Kingdom)

- Building an **end-to-end agentic collections pipeline** (voice, SMS, email) on a multi-tenant core with RBAC, approval rules and compliance-aware audit logging

- Defining the system data model (Tenant → Client → Portfolio → Batch → Account → Case → Evidence) and the escalation rules for disputes, vulnerability and complaints
- Wrote the product requirements and a 10-phase, 90+ ticket implementation plan spanning backend, dashboards, worker daemon and infrastructure

Fullstack Engineer – Beacon Technologies

Apr 2026 – Present

Part-Time (FCT, Nigeria)

- Build secure, auditable payment APIs (JWT, RBAC, PostgreSQL audit trails) and integrate payment processors and hardware devices behind one fault-tolerant service layer
- Ship releases through Docker-based zero-downtime rollouts with structured environment configuration and incident response

Founder & Lead Engineer – LungiLabs

Jul 2025 – Present

Part-Time (Lagos, Nigeria)

- Lead a software studio that has shipped **Vonative** (voice AI) and **Mazamaza** (real-time multiplayer quiz platform, **100+ users**), from architecture and data modelling through deployment and operations
- Own product and technical direction, infrastructure (Docker, Linux VPS, Nginx, CI/CD), payments (Paystack) and client delivery, including a Cloudflare-based agro e-commerce platform

JavaScript Developer (Intern) – Exec200 Systems

Jul 2025 – Dec 2025

Internship (Lagos, Nigeria)

- Built and documented REST services and data integrations for enterprise platforms; improved reliability through regression testing and root cause analysis

Backend Developer (Intern) – AFEX Commodities Exchange

Sep 2023 – Dec 2023

Internship (Ibadan, Nigeria)

- Developed APIs for a high-volume commodities trading platform; optimised SQL schemas and queries and applied Redis caching to cut database load

MENTORING & OUTREACH

AI & Robotics Mentor – AMETTSA

- Mentor students in **artificial intelligence and robotics**, guiding them through core concepts and hands-on projects

TECHNICAL WRITING & COMMUNICATION

- Authored product requirements and architecture documents for Toyiyb, ScholarAI (iterated to v4.0, informed by research into how graduate students search for funding), a real-time computer-vision platform and Sarah AI's autonomous collections system
- Prepared investor and client pitch decks, use-case documents and sector demos (healthcare, banking/fintech) for Vonative, and presented call-automation demos to enterprise prospects

TECHNICAL SKILLS

Programming: Python, TypeScript/JavaScript, SQL

AI & ML: LLMs (OpenAI, Claude, Groq), structured outputs, AI agents and tool calling, embeddings and vector search (pgvector), speech pipelines (ASR/TTS), computer-vision model integration, document parsing (Docling)

Systems: FastAPI, Node.js, PostgreSQL, Redis, Celery, LiveKit, WebSockets, Docker, AWS, Cloudflare Workers, Linux

Engineering Background: Electronics, signals and embedded hardware (NFC/RFID: Beeconpay access-control platform with DESFire EV2 cards and ACR1255U-J1 readers)

HONOURS & AWARDS

- **Presidential Award**, Nigerian University Engineering Students' Association (NUESA), Nov 2024
- **Exemplary Leadership Award**, Lagos State University, Epe Campus, Sep 2024
- **Dangote Foundation Scholar; Federal Government Scholarship (FSB); Lagos State Scholarship Board Scholar; LASU Merit Scholar** (2023–2025)

REFERENCES

Available on request.